

DD_ASR-03 - Complaint

Version: 1.0 Status: Draft for review Bundle: 10_other Audience: customer care developers, ticketing developers, billing developers, work-order developers, frontend developers, QA Related DDs:

- DD_TCK-01 - Ticketing and Case Management
- DD_ASR-01 - Technical Trouble
- DD_ASR-02 - Information Request
- DD_ASR-04 - Service Request
- DD_BIL-02-ADJ-01 - Invoice Adjustments
- DD_WO-01 - Work Order Module
- DD_CUST-INT-01 - Customer Interaction Timeline
- DD_NOT-01 - Notification Service
- FE-APP-01 - BSS Backoffice Web App
- FE-APP-04 - Customer Self Care Frontoffice App

MVP simplification baseline: ASR-03 is implemented as TCK-01 complaint ticket categories, investigation workflow, escalation rules, and optional approval/credit hooks. It is not a separate complaint service, database, or deployment in the first release.

1. Purpose

ASR-03 defines how SOPHIX handles customer complaints.

A complaint is a customer expression of dissatisfaction that needs investigation, accountability, response, and sometimes corrective action. It is more than an information request because the customer says something is wrong, unfair, delayed, repeated, badly handled, or unacceptable.

Examples:

- customer says an invoice charge is wrong;
- customer says the same technical fault keeps returning;
- customer says installation was delayed without explanation;
- customer complains about technician or franchise behavior;
- customer disputes a rejected KYC or order decision;
- customer says the service quality is consistently poor;
- customer asks for compensation or escalation;
- customer reports that support promised an action but it was not done.

This DD does not create a new ticketing engine. Complaints are implemented as configured TCK-01 ticket types/categories with ASR-specific investigation, escalation, resolution-action, and compensation recommendation tables.

2. Ownership Boundary

Concern	Owner
Ticket lifecycle, comments, assignment, SLA, attachments	TCK-01
Complaint category behavior, investigation, root cause, resolution policy	ASR-03 config inside TCK/ASR
Technical diagnosis and repair	ASR-01 / WO-01
Billing correction, credit note, debit note	BIL-02-ADJ-01
Invoice read data	BIL-02-READ-01
Payment read/reconciliation data	PAY-GW-01, BIL-05

Concern	Owner
Subscription action	SUB-LM / SUB-WF owning workflow
Order and installation status	FUL / WO
Customer timeline projection	CUST-INT-01
Customer notifications	NOT-01

Rule: ASR-03 can recommend or request corrective action, but it must not silently mutate subscription, billing, wallet, order, or work-order records. The owning module performs the actual business action through its own API and audit rules.

3. When To Use This Flow

Use ASR-03 when the expected outcome is investigation plus formal response, not only an answer or operational task.

Scenario	Use
Customer asks a neutral question	ASR-02 Information Request
Customer reports service down or degraded now	ASR-01 Technical Trouble
Customer requests an operational action without dissatisfaction	ASR-04 Service Request
Customer says a bill, support action, field visit, staff action, delay, or repeated issue is unacceptable	ASR-03 Complaint
Customer requests money correction or compensation	ASR-03 plus BIL-02-ADJ-01 request if justified
Customer needs repair work	ASR-03 may link/create ASR-01/WO, but WO owns field execution

4. High Level Flow

```

flowchart TD
    A["Customer raises complaint"]
    B["Create TCK complaint ticket"]
    C["Resolve complaint category config"]
    D["Assign complaint owner/queue"]
    E["Start investigation"]
    F["Collect evidence from source modules"]
    G{"Needs field or technical action?"}
    H["Create/link ASR-01 or WO"]
    I{"Needs billing correction or compensation?"}
    J["Create ASR compensation recommendation"]
    K["Request BIL adjustment"]
    L["Record root cause and resolution action"]
    M{"Supervisor review required?"}
    N["Escalate / supervisor review"]
    O["Send customer-visible response"]
    P["Resolve TCK ticket"]

    A --> B --> C --> D --> E --> F --> G
    G -->|Yes| H --> I
    G -->|No| I
    I -->|Yes| J --> K --> L
    I -->|No| L
    L --> M
    M -->|Yes| N --> O
    M -->|No| O
    O --> P
  
```

5. Complaint Types And Categories

ASR-03 primarily uses these TCK ticket types:

TCK type	Use
BILLING_COMPLAINT	Invoice, tax, wallet, payment allocation, discount, or charge dispute.
TECHNICAL_SUPPORT	Repeated or unresolved technical issue where the customer is complaining about handling/quality.
INSTALLATION_FOLLOWUP	Complaint about delayed/missed/poor installation or contractor behavior.
EQUIPMENT_ISSUE	Complaint related to equipment replacement, device quality, recovery, or swap.

TCK type	Use
KYC_SUPPORT	Complaint about rejected, delayed, or mishandled KYC.
GENERAL_INQUIRY	General complaint not fitting a more specific type.
INTERNAL_ESCALATION	Complaint escalated by management or internal control team.

Recommended complaint categories:

Category code	Meaning	Primary source APIs	Default priority
BILLING_DISPUTE	Customer disputes invoice amount, tax, charge, discount, or correction.	BIL-02-READ, BIL-02-ADJ, SIP/DIS if discount related	P2
PAYMENT_DISPUTE	Customer paid but balance/invoice/wallet did not reflect correctly.	PAY-GW, BIL-05, BIL-01-PAY	P2
REPEATED_TECHNICAL_FAULT	Customer reports repeated unresolved technical problem.	ASR-01, WO, provisioning, network tools	P2
SERVICE_QUALITY_COMPLAINT	Customer complains about poor speed, stability, or experience over time.	ASR-01, provisioning, usage/network summaries	P2
INSTALLATION_DELAY_COMPLAINT	Customer complains installation took too long or appointments were missed.	FUL, WO	P2
CONTRACTOR_OR_STAFF_CONDUCT	Customer complains about agent, contractor, technician, or franchise behavior.	EM-01/02, WO, SALES	P1
ORDER_OR_KYC_COMPLAINT	Customer disputes order rejection, KYC rejection, or document handling.	FUL, ILM/KYC	P3
EQUIPMENT_COMPLAINT	Customer complains about faulty, reused, missing, or charged equipment.	OSR, WO, ASR-04	P2
UNFULFILLED_PROMISE	Customer says SOPHIX promised action/credit/callback but did not complete it.	TCK, CUST-INT, NOT	P3
GENERAL_COMPLAINT	Other complaint requiring formal response.	Depends on context	P3

6. ASR-Specific Tables

6.1 asr_complaint_category_config

Stores behavior for each complaint category.

Use this table when the system needs to know the complaint queue, SLA, whether supervisor review is mandatory, which source modules to inspect, and whether compensation may be recommended.

Column	Type	Purpose	Sample
config_id	UUID PK	Config id.	acc-001
operator_code	TEXT	Operator/tenant.	WIK
category_code	TEXT	TCK category code.	BILLING_DISPUTE
default_type_code	TEXT	TCK type code.	BILLING_COMPLAINT
default_priority	TEXT	Initial priority.	P2
default_queue_code	TEXT	Queue that owns first triage.	BILLING_CARE
investigation_required	BOOLEAN	Complaint must have investigation record before closure.	true
supervisor_review_required	BOOLEAN	Supervisor must review resolution before closure.	true
customer_response_required	BOOLEAN	Customer-visible response required before resolution.	true
compensation_allowed	BOOLEAN	Complaint may recommend BIL adjustment.	true

Column	Type	Purpose	Sample
source_modules_json	JSONB	Source modules to inspect.	["BIL-02-READ-01", "BIL-02-ADJ-01"]
allowed_resolution_codes_json	JSONB	Allowed closure codes.	["VALID_COMPLAINT_CORRECTED", "NOT_VALID_EXPLAINED"]
status	TEXT	ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- Unique (operator_code, category_code).
- (operator_code, default_queue_code, status).

6.2 asr_complaint_investigation

Stores the investigation work and facts for a complaint ticket.

Use this table when a care user starts complaint analysis. It keeps the root-cause assessment separate from TCK comments so reporting can answer why complaints happen.

Column	Type	Purpose	Sample
investigation_id	UUID PK	Investigation id.	aci-001
ticket_id	TEXT	TCK ticket id.	tck-20260604-000011
operator_code	TEXT	Operator/tenant.	WIK
category_code	TEXT	Complaint category.	BILLING_DISPUTE
investigation_status	TEXT	OPEN, WAITING_SOURCE, COMPLETED, CANCELLED.	OPEN
assigned_queue_code	TEXT	Queue owning investigation.	BILLING_CARE
assigned_user_id	TEXT NULL	Investigator.	usr-billing-014
source_refs_json	JSONB	Source records reviewed.	[{"module": "BIL-02-READ-01", "ref": "INV-2026-06-000045"}]
customer_claim	TEXT	What the customer is alleging.	Customer says discount was not applied.
findings_summary	TEXT NULL	Internal factual finding.	Discount assignment expired before cycle run.
root_cause_code	TEXT NULL	Root cause taxonomy.	CONFIG_EXPIRED
validity_code	TEXT NULL	VALID, PARTIALLY_VALID, NOT_VALID, INCONCLUSIVE.	VALID
created_by_user_id	TEXT	User/system that opened investigation.	usr-care-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-04T09:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	2026-06-04T11:20:00Z

Recommended indexes:

- Unique (ticket_id).
- (operator_code, category_code, investigation_status).
- (root_cause_code, completed_at DESC).

6.3 asr_complaint_resolution_action

Stores corrective or non-corrective actions decided for the complaint.

Use this table after the investigation determines what SOPHIX should do. A resolution action may be an explanation, apology, work-order follow-up, billing adjustment request, process correction, callback, or no-action decision.

Column	Type	Purpose	Sample
resolution_action_id	UUID PK	Resolution action id.	acra-001
ticket_id	TEXT	TCK ticket id.	tck-20260604-000011
investigation_id	UUID	Investigation id.	aci-001
operator_code	TEXT	Operator/tenant.	WIK

Column	Type	Purpose	Sample
action_type	TEXT	EXPLANATION, APOLOGY, BIL_ADJUSTMENT_REQUEST, WO_FOLLOWUP, PROCESS_CORRECTION, CALLBACK, NO_ACTION.	BIL_ADJUSTMENT_REQUEST
action_status	TEXT	PLANNED, REQUESTED, COMPLETED, REJECTED, CANCELLED.	REQUESTED
owning_module	TEXT NULL	Module that executes action.	BIL-02-ADJ-01
owning_ref_id	TEXT NULL	External action reference.	adj-20260604-000007
customer_visible	BOOLEAN	Can be shown to customer.	true
action_summary	TEXT	Human-readable action.	Requested credit note for incorrect discount charge.
created_by_user_id	TEXT	User/system that created action.	usr-billing-014
created_at	TIMESTAMP	Creation timestamp.	2026-06-04T11:25:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	2026-06-04T13:05:00Z

Recommended indexes:

- (ticket_id, action_status).
- (owning_module, owning_ref_id).

6.4 asr_complaint_escalation

Stores formal escalation of a complaint.

Use this table when the complaint requires supervisor, manager, legal/privacy, finance, NOC, contractor-management, or franchise-management review.

Column	Type	Purpose	Sample
escalation_id	UUID PK	Escalation id.	ace-001
ticket_id	TEXT	TCK ticket id.	tck-20260604-000011
operator_code	TEXT	Operator/tenant.	WIK
escalation_level	TEXT	L2, SUPERVISOR, MANAGER, EXECUTIVE, LEGAL_PRIVACY.	SUPERVISOR
reason_code	TEXT	Why escalation is needed.	COMPENSATION_REVIEW
assigned_queue_code	TEXT	Queue receiving escalation.	CARE_SUPERVISOR
assigned_user_id	TEXT NULL	Specific reviewer.	usr-supervisor-002
status	TEXT	OPEN, APPROVED, REJECTED, CANCELLED.	OPEN
decision_summary	TEXT NULL	Review decision.	Approved customer response and credit request.
created_by_user_id	TEXT	User/system that escalated.	usr-care-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-04T11:30:00Z
decided_at	TIMESTAMP NULL	Decision timestamp.	2026-06-04T12:00:00Z

Recommended indexes:

- (operator_code, status, assigned_queue_code).
- (ticket_id, created_at DESC).

6.5 asr_complaint_compensation_recommendation

Stores a complaint-side recommendation for compensation.

Use this table only when ASR believes a credit/debit adjustment should be proposed. BIL-02-ADJ-01 owns approval and application. ASR does not create credit notes directly.

Column	Type	Purpose	Sample
recommendation_id	UUID PK	Recommendation id.	accr-001

Column	Type	Purpose	Sample
ticket_id	TEXT	TCK ticket id.	tck-20260604-000011
operator_code	TEXT	Operator/tenant.	WIK
reason_code	TEXT	Compensation reason.	WRONG_DISCOUNT_APPLICATION
recommended_direction	TEXT	CREDIT, DEBIT.	CREDIT
recommended_scope	TEXT	BIL adjustment scope: FULL, LINE, AMOUNT.	LINE
recommended_amount	DECIMAL(18,2) NULL	Suggested amount if known.	500.00
currency_code	TEXT	Currency.	KES
source_invoice_id	TEXT NULL	Invoice being corrected.	INV-2026-06-000045
source_line_refs_json	JSONB NULL	Invoice line refs for LINE scope.	[{"lineId":"INVLIN-001","amountOverride":500}]
justification	TEXT	Why compensation is recommended.	Discount assignment was active but not applied in cycle run.
status	TEXT	DRAFT, REQUESTED, APPROVED_BY_BIL, REJECTED_BY_BIL, CANCELLED.	REQUESTED
adjustment_request_id	TEXT NULL	BIL adjustment request id.	adj-20260604-000007
created_by_user_id	TEXT	User/system.	usr-billing-014
created_at	TIMESTAMP	Creation timestamp.	2026-06-04T11:25:00Z

Recommended indexes:

- (ticket_id, status).
- (adjustment_request_id).

6.6 asr_complaint_customer_response

Stores the final response communicated to the customer.

Use this table to preserve the exact customer-visible answer, apology, outcome, and next steps. It complements TCK comments and supports quality review.

Column	Type	Purpose	Sample
response_id	UUID PK	Response id.	acr-001
ticket_id	TEXT	TCK ticket id.	tck-20260604-000011
operator_code	TEXT	Operator/tenant.	WIK
language_code	TEXT	Response language.	en
response_body	TEXT	Customer-visible response.	We found that your discount was not applied correctly. A credit request has been submitted.
response_channel	TEXT	BACKOFFICE, SMS, EMAIL, SELF_CARE, PHONE_SUMMARY.	BACKOFFICE
resolution_code	TEXT	Final resolution code.	VALID_COMPLAINT_CORRECTED
sent_by_user_id	TEXT	User/system.	usr-care-001
sent_at	TIMESTAMP	Response timestamp.	2026-06-04T13:10:00Z

Recommended indexes:

- (ticket_id, sent_at DESC).

7. APIs

ASR-03 APIs run inside the TCK/ASR service. TCK remains source of truth for ticket lifecycle.

7.1 Create Complaint Ticket

Complaints are created through TCK:

```
POST /api/tickets
Idempotency-Key: complaint-20260604-000011
Content-Type: application/json

{
  "operatorCode": "WIK",
  "typeCode": "BILLING_COMPLAINT",
  "categoryCode": "BILLING_DISPUTE",
  "sourceChannel": "BACKOFFICE",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "subject": "Customer disputes invoice discount",
  "description": "Customer says June invoice did not include the approved retention discount.",
  "linkedEntities": [
    { "entityType": "INVOICE", "entityId": "INV-2026-06-000045" }
  ]
}
```

TCK validates type/category, creates the ticket, assigns queue/SLA, and emits generic ticket events.

7.2 Get Complaint Guidance

```
GET /api/asr/complaints/{ticket_id}/guidance
```

Response:

```
{
  "ticketId": "tck-20260604-000011",
  "categoryCode": "BILLING_DISPUTE",
  "defaultQueueCode": "BILLING_CARE",
  "investigationRequired": true,
  "supervisorReviewRequired": true,
  "customerResponseRequired": true,
  "compensationAllowed": true,
  "sourceModules": ["BIL-02-READ-01", "BIL-02-ADJ-01", "SIP-03"],
  "allowedActions": [
    "START_INVESTIGATION",
    "ADD_FINDINGS",
    "CREATE_RESOLUTION_ACTION",
    "RECOMMEND_COMPENSATION",
    "ESCALATE",
    "SEND_CUSTOMER_RESPONSE",
    "RESOLVE"
  ]
}
```

Frontend must use this API to show available actions. Do not hardcode complaint behavior per screen.

7.3 Start Investigation

```
POST /api/asr/complaints/{ticket_id}/investigation
Content-Type: application/json

{
  "customerClaim": "Customer says discount was promised but not applied.",
  "assignedQueueCode": "BILLING_CARE",
  "sourceRefs": [
    { "module": "BIL-02-READ-01", "ref": "INV-2026-06-000045" },
    { "module": "SIP-03", "ref": "discount-assignment-8891" }
  ]
}
```

Creates `asr_complaint_investigation` and moves the TCK ticket to `ASSIGNED` or `WAITING_INTERNAL` depending on source availability.

7.4 Complete Investigation

```
POST /api/asr/complaints/investigations/{investigation_id}/complete
Content-Type: application/json

{
  "findingsSummary": "Discount assignment was active during invoice generation but not included in the billable event cont",
  "rootCauseCode": "BILLING_DISCOUNT_RUNTIME_ERROR",
  "validityCode": "VALID",
  "sourceRefs": [
    { "module": "BIL-02-READ-01", "ref": "INV-2026-06-000045" },
    { "module": "DIS-0P-01", "ref": "discount-runtime-log-7712" }
  ]
}
```

```
}
```

The API records findings. It must not resolve the ticket yet if a required resolution action or customer response is still missing.

7.5 Create Resolution Action

```
POST /api/asr/complaints/{ticket_id}/resolution-actions
Content-Type: application/json

{
  "investigationId": "aci-001",
  "actionType": "BIL_ADJUSTMENT_REQUEST",
  "owningModule": "BIL-02-ADJ-01",
  "customerVisible": true,
  "actionSummary": "Request a credit note for the missing discount amount."
}
```

The action record states what must happen. The owning module still performs the actual action.

7.6 Recommend Compensation

```
POST /api/asr/complaints/{ticket_id}/compensation-recommendations
Content-Type: application/json

{
  "reasonCode": "WRONG_DISCOUNT_APPLICATION",
  "recommendedDirection": "CREDIT",
  "recommendedScope": "LINE",
  "recommendedAmount": 500.00,
  "currencyCode": "KES",
  "sourceInvoiceId": "INV-2026-06-000045",
  "sourceLineRefs": [
    { "lineId": "INVLINE-001", "amountOverride": 500.00 }
  ],
  "justification": "Customer had an active approved discount when invoice was generated."
}
```

If approved by the complaint owner/supervisor, ASR creates a BIL adjustment request:

```
POST /api/adjustments
Content-Type: application/json

{
  "operatorCode": "WIK",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "direction": "CREDIT",
  "scope": "LINE",
  "originalInvoiceId": "INV-2026-06-000045",
  "proposedLineItems": [
    { "parentLineItemId": "INVLINE-001", "amountOverride": 500.00 }
  ],
  "reasonCode": "WRONG_DISCOUNT_APPLICATION",
  "externalCaseRef": "tck-20260604-000011",
  "description": "Credit requested from ASR-03 billing complaint."
}
```

BIL-02-ADJ-01 owns approval, credit/debit generation, wallet/outstanding-balance impact, and financial reporting.

7.7 Escalate Complaint

```
POST /api/asr/complaints/{ticket_id}/escalations
Content-Type: application/json

{
  "escalationLevel": "SUPERVISOR",
  "reasonCode": "COMPENSATION_REVIEW",
  "assignedQueueCode": "CARE_SUPERVISOR",
  "summary": "Supervisor approval required before customer response."
}
```

Escalation moves the ticket to UNDER_REVIEW or keeps it assigned to the escalation queue based on TCK policy.

7.8 Send Customer Response

```
POST /api/asr/complaints/{ticket_id}/customer-response
Content-Type: application/json

{
  "languageCode": "en",
  "responseChannel": "BACKOFFICE",
}
```

```

    "resolutionCode": "VALID_COMPLAINT_CORRECTED",
    "responseBody": "We reviewed your invoice and found that your discount was not applied correctly. A credit request has been submitted."
  }
}

```

The API:

1. stores `asr_complaint_customer_response`;
2. adds a customer-visible TCK comment;
3. optionally calls NOT-01 if a notification must be sent;
4. emits `ComplaintCustomerResponseSent`.

7.9 Resolve Complaint

```

POST /api/tickets/{ticket_id}/resolve
Content-Type: application/json

{
  "resolutionCode": "VALID_COMPLAINT_CORRECTED",
  "resolutionSummary": "Billing complaint validated. Credit request submitted to BIL adjustment workflow."
}

```

TCK resolution must be blocked when required ASR-03 closure conditions are missing.

8. Events

Event	Topic	When emitted
<code>ComplaintCreated</code>	<code>sophix.asr.complaint.created</code>	Complaint ticket is created.
<code>ComplaintInvestigationStarted</code>	<code>sophix.asr.complaint.investigationstarted</code>	Investigation record created.
<code>ComplaintInvestigationCompleted</code>	<code>sophix.asr.complaint.investigationcompleted</code>	Investigation findings completed.
<code>ComplaintResolutionActionCreated</code>	<code>sophix.asr.complaint.resolutionactioncreated</code>	Corrective/non-corrective action created.
<code>ComplaintCompensationRecommended</code>	<code>sophix.asr.complaint.compensationrecommended</code>	ASR recommends BIL adjustment.
<code>ComplaintEscalated</code>	<code>sophix.asr.complaint.escalated</code>	Complaint escalated for review.
<code>ComplaintCustomerResponseSent</code>	<code>sophix.asr.complaint.customerresponsesent</code>	Customer-visible response stored/sent.
<code>ComplaintResolved</code>	<code>sophix.asr.complaint.resolved</code>	Ticket resolved through TCK.

TCK still emits generic ticket events. ASR-03 emits business-specific events for reporting, customer timeline, quality monitoring, and complaint trend analysis.

9. Business Rules

Rule	Description	Reason
R-ASR-COMP-01	Complaint must use configured TCK category.	Avoid unrouteable cases.
R-ASR-COMP-02	Complaint closure requires customer-visible response when category config says so.	Customer must receive formal answer.
R-ASR-COMP-03	Complaint closure requires completed investigation when <code>investigation_required = true</code> .	Prevent unexplained closure.
R-ASR-COMP-04	Supervisor review is required when category config says so, priority is P1, compensation exceeds local threshold, or staff conduct is involved.	Governance.
R-ASR-COMP-05	ASR may recommend compensation but BIL-02-ADJ-01 owns adjustment approval and application.	Financial control.
R-ASR-COMP-06	Customer response must not expose internal staff notes, raw diagnostics, credentials, or blame language.	Privacy and professionalism.

Rule	Description	Reason
R-ASR-COMP-07	Complaint cannot be resolved while required linked WO, BIL adjustment request, or escalation is still open unless resolution policy allows pending follow-up.	Avoid false closure.
R-ASR-COMP-08	Same customer/category/source reference duplicate within configured window must be linked, not treated as unrelated.	Prevent duplicate work.
R-ASR-COMP-09	Complaint validity must be recorded as VALID, PARTIALLY_VALID, NOT_VALID, or INCONCLUSIVE.	Reporting and root-cause analytics.
R-ASR-COMP-10	Ticket resolution must use TCK status APIs.	TCK remains lifecycle source of truth.

10. Common Source APIs

The complaint service gathers facts from owning modules. It must not query another module's database directly.

Complaint category	Source module APIs
Billing dispute	GET /api/invoices/{invoice_id}, GET /api/invoices/{invoice_id}/lines, GET /api/adjustments?externalCaseRef=...
Payment dispute	PAY-GW provider transaction lookup, BIL-01-PAY application lookup, BIL-05 wallet ledger lookup
Repeated technical fault	ASR-01 diagnostic history, TCK linked tickets, WO support history, provisioning observed state
Installation delay complaint	FUL order status, WO installation appointments, contractor assignment from EM-02
Staff/contractor conduct	EM-01/EM-02 actor profile, WO visit records, TCK comments/attachments
KYC/order complaint	FUL KYC step state, ILM customer profile, document metadata from file storage
Equipment complaint	OSR instance registry, RMA/swap history, WO installation/support result
Unfulfilled promise	TCK comments, CUST-INT timeline, NOT delivery records

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice must support:

- complaint creation from customer 360, invoice page, payment page, order page, WO page, and ticket page;
- complaint queue with SLA, priority, category, owner, and escalation state;
- investigation panel with source data links;
- root-cause and validity capture;
- resolution action capture;
- compensation recommendation flow;
- supervisor escalation/review;
- customer-visible response drafting;
- resolve/close through TCK status APIs.

The UI must show which module API each data panel comes from. Example: invoice facts from BIL read API, payment facts from PAY-GW/BIL-05, WO facts from WO-01.

11.2 FE-APP-04 Customer Self-Care

Self-care may allow customers to:

- create selected complaint categories configured with selfcare_allowed in TCK/category policy if added later;

- view complaint ticket status;
- add customer-visible comments or requested documents;
- view final response;
- see linked credit request status only at a customer-safe level.

Self-care must not show internal investigation notes, staff conduct details, source diagnostics, or approval comments.

12. Infra Recommendation

ASR-03 should run inside the TCK service/module for a medium budget deployment. No separate ASR complaint pod is required at the start.

Component	Recommendation
API	Existing TCK API pods expose /api/asr/complaints/....
DB	Existing TCK schema/database with ASR complaint tables.
Workers	Existing TCK worker pool runs complaint SLA, duplicate scan, escalation reminder, and notification retry jobs.
Events	Existing Kafka/MSK topics with ASR complaint topic namespace.
Cache	Optional cache for category guidance only. Do not cache investigation facts as source of truth.

For AWS deployment, use the same ECS/EKS service, RDS PostgreSQL instance/schema, MSK/Kafka cluster, CloudWatch logs/metrics, and IAM policies as TCK. Add CloudWatch alarms for:

- P1 complaint SLA breach;
- open complaints older than configured SLA;
- supervisor review backlog;
- compensation recommendations not linked to BIL adjustment after threshold;
- complaint category spike by operator/region/franchise/contractor.

13. Testing Requirements

13.1 Unit Tests

- Complaint guidance resolves category config.
- Investigation required blocks closure.
- Customer response required blocks closure.
- Supervisor review rule triggers correctly.
- Compensation recommendation maps to BIL adjustment request payload.
- Customer-visible response redaction blocks internal-only terms.
- Duplicate complaint detection links related cases.

13.2 Integration Tests

- Create complaint through TCK and load ASR guidance.
- Start/complete investigation using BIL invoice source refs.
- Create compensation recommendation and BIL adjustment request.
- Escalate to supervisor and approve/reject escalation.
- Send customer response through TCK comment and NOT event.
- Resolve ticket only after closure conditions pass.

13.3 End To End Tests

- Billing dispute validated, credit request created, customer response sent, complaint resolved.
- Complaint not valid, explanation sent, complaint resolved with NOT_VALID_EXPLAINED.
- Staff conduct complaint requires supervisor review before closure.
- Repeated technical fault complaint links ASR-01/WO support history.
- Customer self-care only sees customer-safe complaint status and response.

14. Definition Of Done

- Complaint categories are seeded in TCK.
- ASR complaint config exists per operator/category.
- Complaint APIs support guidance, investigation, resolution action, escalation, compensation recommendation, customer response, and closure validation.
- Closure rules block premature resolution.
- BIL adjustment integration is request-only from ASR and approval/application remains in BIL.
- Backoffice can process the complaint lifecycle end to end.
- Self-care can show customer-safe complaint status and response if enabled.
- Events are emitted for reporting and customer timeline.
- Tests cover billing, technical, staff conduct, and invalid complaint paths.